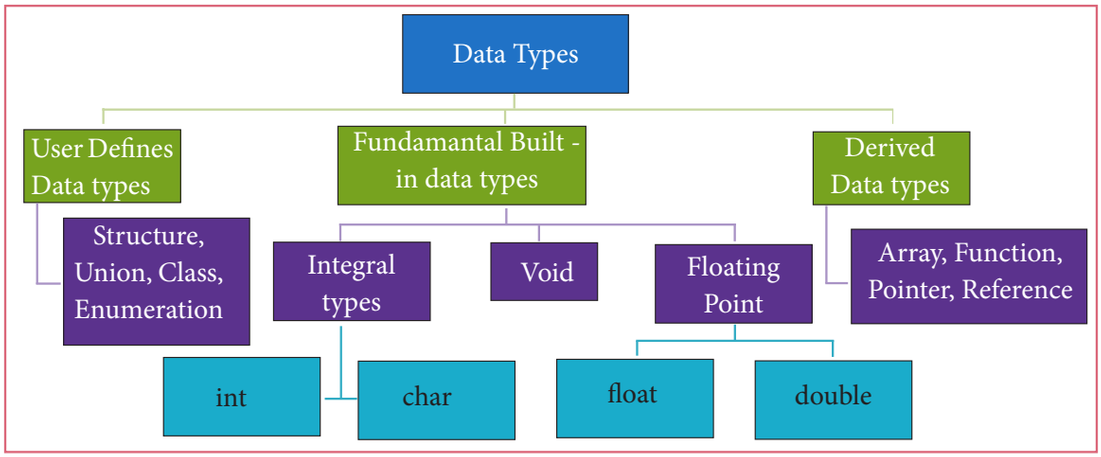

Figure 9.13 Data types in C++

In this chapter, we are going to learn about only the Fundamental data types.

In order to understand the working of data types, we need to know about variables. **T**he variables are the named memory locations to hold values of specific data types. In C++, the variables should be declared explicitly with their data types before they are actually used.

Syntax for declaring a variable:

<data type> <variable name>;

Example:

int num1;

To declare more than one variable which are of the same data type using a single statement, it can be declared by separating the variables using a comma.

Example:

int num1, num2, sum;

For example, to store your computer science marks first you should declare a variable to hold your marks with a suitable data type. Choosing an appropriate data type needs more knowledge and experience. Usually, marks are represented as whole numbers. Thus, the variable for storing the computer science marks should be of integer data type.

Example:

int comp_science_marks;

Now, one variable named **comp_science_marks** is ready to store your marks.

We will learn more about variables later in this chapter.

9.12.1 Introduction to fundamental Data types:

Fundamental (atomic) data types are predefined data types available with C++. There are five fundamental data types in C++: **char, int, float, double** and **void**. Actually, these are the keywords for defining the data types.

(1) int data type:

Integer data type accepts and returns only integer numbers. If a variable is declared as an **int**, C++ compiler allows storing only integer values into it. If you try to store a fractional value in an int type variable it will accept only the integer portion and the fractional part will be ignored.

11th Std Computer Science_EM Chapter 9.indd 133 06-01-2023 15:08:46